

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester I)

Roll No. :S037	Name:HIMANSHU
Class :FYIT	Batch : 2
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Scala For DS-6

Practical 6

Generate a random matrix using breeze and compute its transpose and determinants

Code:

```
import breeze.linalg._
import scala.util.Random
object practical6a extends App{
  val r=3
  val c=3

  val d = Array.fill(r*c)(Random.nextInt(10)).map(_.toDouble)

  val m = new DenseMatrix(r,c,d)

  val trans = m.t

  val d1 = det(m)

  println(s"Original Matrix :\n$m")
  println(s"\nTransposed Matrix :\n$trans")
  println(s"\nDeterminant: $d1")
}
```

Output:

```
Original Matrix :
0.0  1.0  5.0
0.0  0.0  1.0
8.0  4.0  7.0

Transposed Matrix :
0.0  0.0  8.0
1.0  0.0  4.0
5.0  1.0  7.0

Determinant: 8.0

Process finished with exit code 0
```

Himanshu S038